

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations
May-June 2019

Max. Marks:100

Class: TY(IDC)

Name of the Course: Database Management System

Course Code: UCEI605

Duration: 3hrs

Semester: VI

Branch: Comp

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Explain following SQL operations with proper example 1)select 2) <i>Aggregate functions</i> 3)rename 4)Leftouter join 5)Natural join	10
Q 1 (b)	Consider Bank Customer database Customer(name, address, mobileno, dob) Account(accountno, type of account, date of opening the account, balance) Branch(Branch name, address) Transaction(Transactionid, type of transaction, amount) Perform following on above table Create above tables using crate table syntax and insert at least 3 rows in each table(using postgresql) 1 Modify the database so that jack now lives in Mumbai 2 find name of all customer having loan. 3 display the no of Customers having more than 1,0000 in their saving account.	10
Q2 (a)	Write problem definition for Inventory Management system. Draw EER diagram accordingly	10
Q2(b)	What is the need of Database system when file system is available. Explain in detail. <i>OR</i> Draw and Explain the detailed Architecture of Database System with its components	10

Q2 (a)	Compare and contrast Data-Manipulation language and Data-Definition language used for Database. OR Write short note on any two 1) Data independence 2) Data Dictionary 3) Database administrator	10
Q3. (b)	Explain 1NF, 2NF, and 3NF with example. OR Explain Multivalued dependency with example. How it is normalized	10
Q4 (a)	Explain ACID properties of transactions with example. OR Explain Two-Phase Locking Protocol used for serializability	10
Q4 (b)	Consider the three transaction T1,T2,T3, and schedule S given below, draw the precedence graph for S and state whether given schedule is conflict serializable or not, if the schedule is conflict serializable write down the equivalent serial schedule. T1:r1(x);r1(z);w1(x); T2:r2(z);r2(y);w2(z);w2(y); T3:r3(x);r3(y);w3(y); S:r1(x);r2(z);r1(z);r3(x);r3(y);w1(x);w3(y);r2(y);w2(z);w2(y)	10
Q5 (a)	Explain the importance of " JDBC ODBC connections for Database" in a two tier architecture.	10
Q5(b)	Define Normalization . Explain Lossless join decomposition and Dependency Preserving decomposition with example	10

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End Semester Exam

May - Jun 2019

Max. Marks:100

Class: T.Y.B.Tech.

Name of the Course: Mobile Cellular and Ad Hoc Networks

Course Code: UCEC604

Duration:3Hrs

Semester: VI

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Describe the functions of the MS and SIM. Why does GSM separate the MS and SIM? How and where is user related data represented/stored in GSM system? OR How is synchronization achieved in GSM? Who is responsible for Synchronization and why is it so important?	10 (4+2+4) 10 (6+2+2)
Q 1 (b)	What types of Handover can occur in GSM? Explain the typical signal flow during handover with the help of message flow diagram for intra-MSC handover?	10 (4+6)
Q2 (a)	Why is new infrastructure needed for GPRS, but not for HSCSD? Which components are new and what is their purpose?	10 (2+8)
Q2 (b)	How does UTRA-FDD counteract the near-far effect? Why is this not a problem in GSM? OR Explain with the help of diagram, UMTS core network with 3G RNS and 2G BSS.	10 (6+4) 10
Q3 (a)	Compare the overhead introduced by all three medium access schemes (DCF using CSMA/CA, DCF with RTS/CTS, PCF with polling) and the resulting performance at zero load, light load, high load of the medium in IEEE 802.11. How does number of collisions increase with the number of stations trying to access the medium?	10 (6+4)
Q3 (b)	Explain IEEE 802.11 b PHY layer with the help of packet formats. OR Explain ,How QoS provided in 802.16 WiMax.	10 10
Q4 (a)	Explain AODV routing technique. And Explain expanded ring search technique. How it affects performance?	10 (5+3+2)
Q4 (b)	Explain Optimized Link State Routing protocol with its advantages and disadvantages. OR Explain the route discovery process in DSR with example. Discuss the use of route caching in DSR.	10 (6+4) 10 (6+4)

Q5 (a)	Discuss the issues that make wireless sensor networks different than other ad-hoc wireless networks.	10
Q5 (b)	Write short note on VANET.	10

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End Semester Examinations

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Class: ~~T.Y.~~ B. Tech

Name of the Course: Digital signal and Image Processing

Course Code: UCEC603

Duration: 3Hrs

Semester: VI

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks																		
Q 1 (a)	1) State the condition of stability of LTI system and determine for the given time system $h(n) = (0.3)^n u(n) + 5\delta(n)$ is stable or not 2) For the given causal sequences $x(n) = \{8, 9, 2, 3\}$ and $h(n) = \{4, 3, 6\}$. Find the cross correlation. OR Check whether following system $y(n) = 2x(n-1) + x(2n)$ is : 1. Linear or non-Linear 2. Causal or non-causal 3. Time variant or Time invariant 4. Static or Dynamic	10M																		
Q 1 (b)	Given <table border="1"><tr><td>13</td><td>54</td><td>12</td></tr><tr><td>13</td><td>11</td><td>57</td></tr><tr><td>11</td><td>10</td><td>12</td></tr></table> Find 3 bit IGS coded image and calculate compression factor ,BPP and MSE	13	54	12	13	11	57	11	10	12	10M									
13	54	12																		
13	11	57																		
11	10	12																		
Q2 (a)	Perform histogram equalization and draw new equalized histogram of the following image data <table border="1"><tr><td>Grey Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No of Pixel</td><td>70</td><td>100</td><td>40</td><td>80</td><td>60</td><td>40</td><td>8</td><td>2</td></tr></table>	Grey Level	0	1	2	3	4	5	6	7	No of Pixel	70	100	40	80	60	40	8	2	10M
Grey Level	0	1	2	3	4	5	6	7												
No of Pixel	70	100	40	80	60	40	8	2												

Q2 (b)	Find DCT of the given image using matrix multiplication method: <table><tr><td>2</td><td>4</td><td>4</td><td>2</td></tr><tr><td>4</td><td>6</td><td>8</td><td>3</td></tr><tr><td>2</td><td>8</td><td>10</td><td>4</td></tr><tr><td>3</td><td>8</td><td>6</td><td>2</td></tr></table>	2	4	4	2	4	6	8	3	2	8	10	4	3	8	6	2	10M
2	4	4	2															
4	6	8	3															
2	8	10	4															
3	8	6	2															
Q3 (a)	Explain any five zero memory point operations.	10M																
Q3 (b)	Write 8X8 Hadamard matrix and its signal flow graph. Using Butterfly diagram, compute Hadamard transform for $X(n)=\{1,0,1,0,0,1,1,0\}$	10M																
Q4 (a)	Explain Morphological operation Opening and Closing <i>OR</i> <i>Explain 4, 8 and m connectivity of a pixel with examples</i>	10M																
Q4 (b)	Explain Hough transform with suitable example <i>OR</i> Explain in detail Moments , Normalised moments and central moments.	10M																
Q5 (a)	What is image segmentation? Explain the following methods of image segmentation. 1) Region growing 2) Split and Merge <i>OR</i> What are different types of redundancies in digital image? Explain in detail giving example of each.	10M																
Q5 (b)	Write short notes on <i>(any two)</i> 1) Homomorphic filter 2) Fidelity criteria 3) <i>Haar Transform</i> <i>OR</i> Explain : The first difference makes the chain code invariant to rotation	10M																

20-5-2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
May - Jun 2019

Max. Marks:100

Class: T.Y.B.Tech.

Name of the Course: System Programming and Compiler Construction

Course Code: UCEC602

Duration:3Hrs

Semester: VI

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain the working of single pass assembler with example.	10
Q 1 (b)	For the following program: 1. Write the expanded assembly language programming 2. Draw MDT table after macro processing 3. Draw MNT table after macro processing 4. Write ALA generated in pass I and pass II Program: <pre> MACRO XYZ &A ST 1,&A MEND MACRO MIT &Z AR 4, &Z ST &Z, ALL MEND PROG START USING *,15 MIT YALE ST 2,3 XYZ ALL YALE EQU 5 ALL DC F'3' END </pre>	10
Q2 (a)	Explain designing of pass I of Dynamic linking loader. What is the order of RLD, ESD, TXT and END cards in the object deck? Why are they in this order?	10 (6+1+3)
Q2 (b)	Differentiate between (any TWO): 1. Image builder and Linkage editor 2. Compiler and Interpreter 3. System program and Application program	10

Q3 (a)	Consider the grammar $S \rightarrow AS \mid b$ $A \rightarrow SA \mid a$ - Construct the collection of set of LR(0) items for this grammar - Construct the goto graph of LR(0) item sets - Construct the parsing table using SLR Algorithm - Show all moves allowed by table from (c) on the input abab using LR parsing algorithm. OR Show that the following grammar is LR(1) but not LALR(1) $S \rightarrow Aa \mid bAc \mid dc \mid bda$ $A \rightarrow d$	15
Q3 (b)	Show that the following grammar is LL(1) $S \rightarrow AaAb \mid BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$	5
Q4 (a)	Construct the DAG and identify the value numbers for the subexpressions of the following expression in value-number method, Assuming + associates from the left : $a+a+(a+a+a+(a+a+a+a))$	10 (5+5)
Q4 (b)	Explain storage allocation strategies.	10
Q5 (a)	Translate the executable statements of following statements into three address code. <pre> int i,x; int a[10]; i=1; while(i<=10){ if(x<1){ a[i]=0; } i=i+1; } </pre> OR Explain issues in the design of code generator.	10
Q5 (b)	Explain any two approaches used in garbage collection algorithm in JAVA. OR Write short note on - Lex and Yacc - Peephole optimization techniques (any two with proper example)	10

15.05.2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations

May-June 2019

Max. Marks: 100

Class: Third Year

Name of the Course: Software Engineering

Branch: Comp

Course Code: UCEI602

Duration: 3 hours

Semester: VI

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Explain Capability Maturity Model in detail OR Explain version control and change control in SCM	(10 M) (10 M)
Q 1 (b)	What are software risks? Discuss RMMM with an example. OR What is agile methodology? Discuss its principles and explain any one agile process model	(10 M) (10M)
Q2 (a)	Compute the cyclomatic complexity for the following code after drawing the flow graph <pre> int BinSearch (char *item, char *table[], int n) { int bot = 0, mid, cmp; int top = n - 1; while (bot <= top) { mid = (bot + top) / 2; if (table[mid] == item) return mid; else if (compare(table[mid], item) < 0) top = mid - 1; else bot = mid + 1; } return -1; } </pre> OR	(10 M)

	Explain the concept of equivalence partitioning in detail	(10 M)
Q2 (b)	Explain the concept of OO testing in detail.	(10 M)
	OR	
	Compare and contrast between white box testing and black box testing	(10 M)
Q3 (a)	What is SRS document? Build an SRS for Online Hospital system	(10 M)
Q3 (b)	Draw the use case diagram and activity diagram for the above system	(10 M)
Q4 (a)	What do you understand by software maintenance? Explain the different types of software maintenance.	(10 M)
Q4 (b)	Explain the concepts of reverse engineering and re-engineering.	(10 M)
Q5 (a)	Explain the various software design concepts?	(10 M)
	OR	
	What is user interface design process? Explain with one example	(10 M)
Q5 (b)	Define coupling and cohesion. Discuss the various types of coupling and cohesion	(10 M)

15.5.2019 (E)

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End Semester Exam

May 2019

Max. Marks: 100

Class: TY

Name of the Course: Management Information Systems

Branch: COMP

Duration: 03 Hrs.

Semester: VI

Course Code: UCEI601

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary and state it clearly

Question No.		Max. Marks
Q1(A)	Discuss the need for MIS as per business and Information Systems.	10
	OR	
	What are management challenges of Information Technology? Explain with a suitable example.	
Q1(B)	Explain BPR. Elaborate on role played by IT in BPR.	10
Q2 (A)	Differentiate between MIS Plan Vs Business Plan. With an example explain development of long range plans for MIS.	10
Q2(B)	What are different ERP benefits, costs and causes of failure?	10
	OR	
	Explain Supply Chain life cycle. What are different benefits and challenges of SCM?	
Q3(A)	Write short note on any 2: 1. DSS 2. EIP 3. KMS	10
Q3(B)	With an example case study explain different levels of decision making.	10
Q4(A)	Explain Organizational Planning with neat diagram.	10
Q4(B)	What are different Intranet Enterprise Portal challenges?	10
	OR	
	Define Change Management. What are key dimensions of Change Management? Explain.	
Q5	Attempt any 4: 1. Explain key elements of Benchmarking. 2. Explain steps of Benchmarking process. 3. Explain PERT using a suitable example. 4. Explain CPM approach in detail. 5. Differentiate between PERT and CPM. 6. Explain SDLC model with neat diagram.	20